LINUX FUNDAMENTALS – WEEK 1 LECTURE NOTES

Introduction to Linux

Linux is an open-source, Unix-like operating system developed by Linus Torvalds. It is widely used in servers, cloud computing, embedded systems, cybersecurity, and software

development.

Features of Linux

■ **Open Source**

■ **Multiuser**

■ **Multitasking**

■ **Secure**

■ **Portable**

■ **Stable**

■ **Command Line Support**

■ **GUI Support**

Learning Objectives

■ **Understand Linux architecture.**

■ **Navigate the Linux filesystem.**

■ **Use command-line tools effectively.**

■ **Manage files and directories.**

■ **Configure Linux systems.**

■ **Install and manage software packages.**

■

Linux Philosophy and Concepts

Core Concepts

Open Source Software

The source code is publicly available and can be modified or distributed by anyone.

Linux Architecture

Hardware → Kernel → Shell → User Applications

Kernel

The kernel is the core component of Linux responsible for:

- **Process Management**
- **Memory Management**
- **Device Management**
- **File System Management**

Shell

The shell acts as an interface between the user and the kernel.

Examples:

- **Bash**
- **Zsh**
- **Fish**

Linux Principles

- **Everything is a File**
- **Small Modular Tools**
- **Command-Line Efficiency**
- **User Permissions and Security**
- **Flexibility and Portability**

Linux Basics and System Startup

Linux Boot Process

1. Power On
2. BIOS/UEFI Initialization
3. Bootloader (GRUB)
4. Kernel Loading
5. System Initialization
6. Service Startup
7. User Login

Important Terms

BIOS

Basic Input Output System used to initialize hardware during startup.

UEFI

Modern replacement for BIOS with enhanced features.

Bootloader

Loads the Linux kernel into memory.

Example: GRUB (Grand Unified Bootloader)

Daemon

Background process running continuously in Linux.

Examples:

■ sshd

■ cron

■ systemd

Startup Commands

Check system information:

uname -a

Check kernel version:

uname -r

Shutdown system:

sudo shutdown now

Restart system:

sudo reboot

Graphical Interface

Desktop Environment

A Desktop Environment provides graphical interaction with the system.

Popular Desktop Environments:

- **GNOME**
- **KDE Plasma**
- **XFCE**
- **Cinnamon**

Features

- **Desktop**
- **File Manager**
- **Panels**
- **Workspaces**
- **Application Launcher**
- **Settings Manager**

Desktop Operations

- **File Browsing**
- **Window Management**
- **Display Configuration**
- **Theme Customization**

■ Workspace Management

System Configuration Using GUI

User Management

Add User:

sudo adduser username

Delete User:

sudo deluser username

Network Configuration

View IP Address:

ip a

Check Hostname:

hostname

Change Hostname:

sudo hostnamectl set-hostname newname

System Monitoring

View Running Processes:

top

View Memory Usage:

free -h

View Disk Usage:

df -h

Common Linux Applications

Web Browsers

■ **Firefox**

■ **Chromium**

Office Applications

■ **LibreOffice**

Media Players

■ **VLC Media Player**

Text Editors

■ **Nano**

■ **Vim**

■ **Gedit**

Development Tools

■ **VS Code**

■ **Eclipse**

■ **IntelliJ IDEA**

Compression Utilities

■ **zip**

■ **unzip**

■ **tar**

Command Line Operations

Terminal Basics

Open Terminal:

Ctrl + Alt + T

Current User:

whoami

Current Directory:

pwd

Display Date:

date

Display Calendar:

cal

Clear Terminal:

clear

Directory Navigation

List Files and Directories:

ls

Detailed Listing:

ls -l

Show Hidden Files:

ls -la

Change Directory:

cd directory_name

Move to Home Directory:

cd ~

Move Back One Directory:

cd ..

Move to Root Directory:

cd /

Print Working Directory:

pwd

File and Directory Management

Create File

touch file.txt

Create Directory

mkdir myfolder

Remove Empty Directory

rmdir myfolder

Remove File

rm file.txt

Remove Directory Recursively

rm -r myfolder

Copy File

cp source.txt destination.txt

Move/Rename File

mv oldname.txt newname.txt

Viewing File Contents

Display Entire File:

cat file.txt

View File Page by Page:

less file.txt

View First 10 Lines:

head file.txt

View Last 10 Lines:

tail file.txt

Hard Links and Symbolic Links

Hard Link

ln file1 hardlink

Symbolic Link

ln -s file1 symlink

List Inode Numbers:

ls -li

Searching for Files

Locate Command

Search File:

locate filename

Update Database:

sudo updatedb

Find Command

Find File by Name:

find /home -name file.txt

Find Directories:

find /home -type d

Find Large Files:

find / -size +10M

Find Files Modified in Last Day:

find /home -mtime -1

Wildcards

All Text Files:

ls *.txt

Files Starting with “a”:

ls a*

Single Character Match:

ls file?.txt

Input and Output Redirection

Redirect Output:

ls > output.txt

Append Output:

ls >> output.txt

Redirect Input:

cat < input.txt

Redirect Errors:

command 2> error.txt

Pipes

A pipe passes output from one command as input to another.

Example:

ls -l | less

Count Files:

ls | wc -l

Search Text:

cat file.txt | grep Linux

User Privileges and Sudo

Execute Command as Administrator:

sudo command

Switch User:

su username

Switch to Root:

sudo -i

Package Management

APT Package Manager (Ubuntu/Debian)

Update Package List:

sudo apt update

Upgrade Installed Packages:

sudo apt upgrade

Install Package:

sudo apt install package_name

Remove Package:

sudo apt remove package_name

Search Package:

apt search package_name

List Installed Packages:

apt list --installed

Key Takeaways

■ Learned Linux architecture and operating principles.

■ Understood Linux boot process and startup sequence.

■ Explored graphical desktop environments.

■ Configured Linux systems using GUI tools.

■ Practiced command-line operations.

■ Managed files and directories efficiently.

■ Used links, wildcards, pipes, and redirection.

■ Installed and removed software packages.

■ Gained hands-on experience through laboratory exercises.

■ Created a GitHub repository to document internship activities and learning

progress.